TS-006
CS-36

Education to Eternity

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

An Autonomous Institute, with NAAC "A" Grade

Basic Science & Humanities Department

Session: 2023-24 (EVEN Semester)

MSE EXAMINATION

Course: B. Tech. Subject: Linear Algebra and Complex Analysis
Subject Code:CS/AI/DS2T001
Year/Semester:IIndSemester (First Year)
Max Marks: 20 **Date :- 16/04/2024** **Duration:- 1 Hr**

Instructions to the Students:
1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO1.Describe concept of complex numbers, integral calculus & multiple integrals, probability and distribution theory.
CO2 Illustrate the concept of complex numbers, integral calculus & multiple integrals, probability and distribution theory by using examples.
CO3 Apply the knowledge of complex numbers, integral calculus & multiple integrals, probability and distribution theory to solve the engineering problems.
CO4 Analyze the problems and results of complex numbers, integral calculus & multiple integrals, probability and distribution theory to solve the real world problems.
CO5 Evaluate the problems by using complex numbers, integral calculus & multiple integrals, probability and distribution theory.
CO6 Create the methods or model by using complex numbers, integral calculus & multiple integrals, probability and distribution theory.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Matrix.	L1/CO1	1
Q.2	Write down the formula of Integrating factor for $\frac{dy}{dx} + Py = Q$.	L1/CO1	1
Q.3	Define order of differential equation.	L2/CO2	1
Q.4	Write down the formula for solution of exact differential equation.	L1/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Interpret Integrating factor of $\frac{dy}{dx} - \frac{3}{x}y = \frac{x-1}{x}$. $\frac{1}{x^3}$	L 2/CO2	2
Q.6	Interpret whether the equation $\frac{2y}{x}\,dx + (2\log x - y)\,dy = 0$ is <u>exact</u> or not.	L 2/CO2	2
Q.7	Find Eigen values of $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$. 8, 2, 2	L2/CO2	2
Q.8	InterpretRank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$. 1	L2/CO2	2

SECTION C

1. **Solve any two questions.**

Q.9	Identify$B^{-1}AB$ if $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$.	L3/CO3	4
Q.10	Solve $(x+1)\frac{dy}{dx} - 2y = (x+1)^4$ $\frac{y}{(x+1)^2} = \log(x+1) + c$	L3/CO3	4
Q.11	Analyze $(2x - y + 1)dx - (x - 2y + 1)dy = 0$ $(x^2 - x) + (y^2 - y) + c$	L4/CO4	4